数据格式

数据标准格式：

上传的文件夹：

yyyy_mm_dd/username/robot/task/04:d/episode_xxxx

yyyy_mm_dd 以采集的时间为准

键名

没有的键不保存文件夹

observation.state.joint_position：本体关节参数（双臂采用左+右拼接） 弧度

- timestamp, left_j1, left_j2 .... , left_gripper, right_j1, ...right_gripper

observation.state.eef_pose：10/20维（xyz 3维 + 6D旋转 6维 + gripper 1维（单位米））

- timestamp, left_x, left_y, left_z, left_r1, ..., left_r6, left_gripper, right_x....,...

observation.state.joint_velocity：本体关节速度（双臂采用左+右拼接）

- timestamp, left_v1, left_v2, ..., right_v1,

observation.state.gripper：1/2维（单位米）

- timestamp, left_gripper, right_gripper

actions先于observation一帧，最后一帧复制

actions.eef_pose：10/20维，本体末端参数（双臂采用左+右拼接）（只有umi存，作为state向后偏移的一帧）

- timestamp, left_x, left_y, left_z, left_r1, ..., left_r6, left_gripper, right_x.... , ...

actions.joint_position：本体关节控制参数（双臂采用左+右拼接，遥操和spacemouse存）-> 更正：Gello 控制参数保存入 actions.joint_position，SpaceMouse 控制参数保存入 actions.eef_pose。

- timestamp, left_j1, left_j2 .... , left_gripper, right_j1, ...right_gripper

observation.image.third_view：(H, W, 3)

observation.image.second_third_view：(H, W, 3)

observation.image.left_wrist_view：(H, W, 3)

observation.image.right_wrist_view：(H, W, 3) observation.image.left_wrist_right_tactile: (H, W, 3)

observation.image.left_wrist_left_tactile: (H, W, 3)

observation.image.right_wrist_right_tactile: (H, W, 3)

observation.image.right_wrist_left_tactile: (H, W, 3)

meta/info.json

Code block


```
{
  "session_id": "5f21de3f-3277-41d7-bbca-32c9a4a7fa09",
  "recording_session_id": "d031a3e1-1948-4a74-ae51-922a8212cfb0",
  "task_id": "cee586ae-8c46-4a3d-b2ae-ed7c046ae2eb",
  "task_key": "pick_cube",
  "task_title": "拿起方块",
  "language_instruction": "",
  "robot": "arx5",
  "robot_sn": "ARX5 Bimanual",
  "username": "lirui",
  "episode_index": 0,
  "episode_id": "episode_0000",
  "total_frames": 1702,
  "fps_actual": 29.9,
  "fps_config": 30,
  "start_time": "2026-04-08T14:35:00.428086+00:00",
  "end_time": "2026-04-08T14:35:57.307110+00:00",
  "duration_seconds": 56.88,
  "created_at": "2026-04-08T14:35:00.428086+00:00",
  "cameras": {
    "observation.image.third_view": {
      "width": 640,
      "height": 480,
      "fps": 30,
      "type": "realsense",
      "device": "346522070312"
    },
    "observation.image.left_wrist_view": {
      "width": 640,
      "height": 480,
      "fps": 30,
      "type": "realsense",
      "device": "409122274673"
    },
    "observation.image.right_wrist_view": {
      "width": 640,
```

```json
        "height": 480,
        "fps": 30,
        "type": "realsense",
        "device": "409122273847"
      }
    },
    "modalities": {
      "observation.image.third_view": {
        "type": "video",
        "frames": 1702,
        "hz_nominal": 30
      },
      "observation.image.left_wrist_view": {
        "type": "video",
        "frames": 1702,
        "hz_nominal": 30
      },
      "observation.image.right_wrist_view": {
        "type": "video",
        "frames": 1702,
        "hz_nominal": 30
      },
      "observation.state.joint_position": {
        "type": "csv",
        "rows": 1702,
        "hz_nominal": 30
      },
      "observation.state.joint_velocity": {
        "type": "csv",
        "rows": 1702,
        "hz_nominal": 30
      },
      "observation.state.eef_pose": {
        "type": "csv",
        "rows": 1702,
        "hz_nominal": 30
      },
      "observation.state.gripper": {
        "type": "csv",
        "rows": 1702,
        "hz_nominal": 30
      },
      "actions": {
        "type": "csv",
        "rows": 0,
        "hz_nominal": 30
      }
```

```
  },
  "frame_integrity": {
    "observation.image.third_view": {
      "total_drops": 1,
      "max_consecutive_drops": 1
    },
    "observation.image.left_wrist_view": {
      "total_drops": 28,
      "max_consecutive_drops": 2
    },
    "observation.image.right_wrist_view": {
      "total_drops": 32,
      "max_consecutive_drops": 2
    }
  },
  "nas_uploaded": false,
  "quality": {
    "labels": [
      "完全正常"
    ]
  }
}
```

Following format contents are all DEPRECATED! They are no longer in use!

ARX

Code block

```json
{
  "observation": {
    "state": "float32[14]",
    "ee_pose": "float32[20]",
    "image/top": "video frame reference",
    "image/left_wrist": "video frame reference",
    "image/right_wrist": "video frame reference",
    "image/tactile_ll": "video frame reference",
    "image/tactile_lr": "video frame reference",
    "image/tactile_rl": "video frame reference",
    "image/tactile_rr": "video frame reference"
  },
  "action": "float32[14]",
  "discount": 1.0,
  "reward": null,
  "is_first": false,
  "is_last": false,
  "is_terminal": false,
  "language_instruction": "pick place",
  "timestamp": 1.2333333,
  "timestep": 37
}
```

Flexiv

Code block

```
1. meta/info.json — Episode 元数据

 {
   "episode_id": "episode_000005",         // episode 文件夹名
   "task": "st",                           // 任务标识（task_prefix）
   "robot": "flexiv_rizon4",               // 机器人型号
   "operator": "stress_test",              // 操作员用户名
   "scene": "",                            // 场景描述（预留）
   "start_time_utc": "2026-03-26T03:36:57.118651Z",  // 开始时间（UTC）
   "end_time_utc": "2026-03-26T03:36:58.909696Z",    // 结束时间（UTC）
   "duration_ns": 1791044950,              // 持续时长（纳秒）
   "success": true,                        // 是否成功（false=标记了异常）
   "language_instruction": "st",           // 自然语言任务描述
   "notes": "",                            // 异常原因（success=false时填写）
   "fps": 30,                             // 采集帧率
   "total_frames": 36                      // 总帧数
   // 可选字段:
   // "discard": true                      // 选择"完全错误"时标记，表示废弃不可用
 }
```

Code block

```
2. meta/episode.json — 模态摘要

 {
   "modalities": {
     "joint_position": { "type": "csv", "rows": 36, "hz_nominal": 30 },
     "joint_velocity": { "type": "csv", "rows": 36, "hz_nominal": 30 },
     "eef_pose":       { "type": "csv", "rows": 36, "hz_nominal": 30 },
     "gripper":        { "type": "csv", "rows": 36, "hz_nominal": 30 },
     "action":         { "type": "csv", "rows": 36, "hz_nominal": 30 },
     "wrist_left":     { "type": "video", "frames": 36, "hz_nominal": 30 },
     "main_cam":       { "type": "video", "frames": 36, "hz_nominal": 30 },
     "third_view_left":  { "type": "video", "frames": 36, "hz_nominal": 30 },
     "third_view_right": { "type": "video", "frames": 36, "hz_nominal": 30 }
   }
 }
```

Code block

```
3. meta/modality_config.json — 硬件配置

 {
   "cameras": {
     "wrist_left": {
       "model": "USB Camera",
```

```
        "resolution": [640, 480],
        "codec": "h264",
        "fps_target": 30
      },
      "main_cam": {
        "model": "RealSense",
        "resolution": [640, 480],
        "codec": "h264",
        "fps_target": 30,
        "serial_number": "409122272546"     // RealSense 有序列号
      }
      // ... 其他相机
    },
    "robot": {
      "model": "flexiv_rizon4",
      "dof": 7,
      "joint_names": ["joint_1", ..., "joint_7"],
      "joint_limits_lower": [-2.7925, -2.2689, -2.7925, -2.2689, -2.7925, -2.2689, -2.7925],
      "joint_limits_upper": [2.7925, 2.2689, 2.7925, 2.2689, 2.7925, 2.2689, 2.7925],
      "gripper_type": "flexiv"
    },
    "action_space": {
      "type": "absolute_tcp_6d",
      "dim": 10,
      "layout": ["tcp.x", "tcp.y", "tcp.z", "tcp.r1", ..., "tcp.r6", "gripper.pos"],
      "hz": 30
    }
  }
```

Code block

```

补充：Record 级别文件

metadata.json（session 级别，非 episode 内）：
{
  "metadata_version": 1,
  "task_key": "clip_chips",
  "task_id": "clip_chips",
  "task_title": "clip_chips",
  "username": "test",
  "start_time": "2026-03-26 19:08:05",     // 本地时间
  "duration_seconds": 100.21,
```

```
    "fps": 30.0,
    "total_frames": 1949,
    "total_episodes": 1,
    "file_count": 26,
    "total_size": 305351915,
    "nas_uploaded": false,
    "anomaly_flag": false,                    // 是否有异常 episode
    "anomaly_reason": null,                   // 异常摘要
    "anomaly_episodes": []                    // [{episodeIndex: N, reason: "..."}, ...]
  }
```

Code block

```
{root}/006/                               # 提前设置好录制条数，一次录制的数据都会存到一个文件夹下，episode从0开始
  ├── episode_000000/                     # 每条 episode 独立文件夹
  │   ├── action/
  │   │   └── data.csv                     # timestamp_ms, joint_1.pos ~ joint_7.pos, gripper.pos
  │   ├── joint_position/
  │   │   └── data.csv                     # timestamp_ms, j1 ~ j7 (机器人实际关节位置)
  │   ├── joint_velocity/
  │   │   └── data.csv                     # timestamp_ms, j1_vel ~ j7_vel
  │   ├── eef_pose/
  │   │   └── data.csv                     # timestamp_ms, x, y, z, r1~r6, gripper (末端位姿+夹爪)
  │   ├── gripper/
  │   │   └── data.csv                     # timestamp_ms, width, force
  │   ├── wrist_left/
  │   │   ├── video.mp4                    # 视频流
  │   │   └── timestamps.txt               # 每帧真实采集时间 (ms, 相对第一帧)
  │   ├── main_cam/
  │   │   ├── video.mp4
  │   │   └── timestamps.txt
  │   ├── third_view_left/
  │   │   ├── video.mp4
  │   │   └── timestamps.txt
  │   ├── third_view_right/
  │   │   ├── video.mp4
  │   │   └── timestamps.txt
  │   └── meta/
  │       ├── info.json                    # episode 元数据 (fps, total_frames, duration, success 等)
  │       ├── episode.json                 # 各模态的帧数/类型摘要
```

```
│       └── modality_config.json     # 相机/机器人/动作空间配置
├── episode_000001/
│   └── ...
└── episode_000049/
    └── ...

```

Code block

```
各 CSV 的 key:


┌────────────────────────────┬──────────────────────────────────────────────────────────────┐
│           文件             │                              列                              │
├────────────────────────────┼──────────────────────────────────────────────────────────────┤
│ action/data.csv            │ timestamp_ms, joint_1.pos, joint_2.pos, ..., joint_7.pos, gripper.pos │
├────────────────────────────┼──────────────────────────────────────────────────────────────┤
│ joint_position/data.csv │ timestamp_ms, j1, j2, ..., j7                                   │
├────────────────────────────┼──────────────────────────────────────────────────────────────┤
│ joint_velocity/data.csv │ timestamp_ms, j1_vel, j2_vel, ..., j7_vel                       │
├────────────────────────────┼──────────────────────────────────────────────────────────────┤
│ eef_pose/data.csv          │ timestamp_ms, x, y, z, r1, r2, r3, r4, r5, r6, gripper           │
├────────────────────────────┼──────────────────────────────────────────────────────────────┤
│ gripper/data.csv           │ timestamp_ms, width, force                                      │
└────────────────────────────┴──────────────────────────────────────────────────────────────┘

- action = teleop 发出的指令（关节目标位置）
```

```
    - joint_position = 机器人实际关节位置（反馈值）
    - joint_velocity = 机器人实际关节速度
    - eef_pose = 末端执行器位姿（10 维：xyz + 旋转矩阵前两行 + 夹爪）
    - gripper = 夹爪宽度和力
```

ALOHA